Multimodal Machine Learning for Alzheimer's Disease Classification Using Clinical Data and MRI Scans from OASIS

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Abstract
This project develops a multimodal machine learning approach to classify Alzheimer's disease using both clinical and imaging data. The study aims to evaluate whether combining clinical and demographic variables with MRI-derived features improves classification performance compared to using either modality alone. To address this, MRI images will be processed using a convolutional neural network (CNN) to extract feature representations, which will then be combined with clinical patient data. The resulting feature vector will be used as input to a multilayer perceptron (MLP) classifier to generate final predictions. Model performance will be compared across clinical-only, MRI-only, and multimodal approaches using standard evaluation metrics. It is expected that the multimodal model will outperform single-modality models, as clinical and imaging data provide complementary information about disease status.

Introduction
Alzheimer's disease (AD), a neurodegenerative disease and common form of dementia, affects one's ability of thinking, remembering, and reasoning. Approximately, more than 55 million people in the world live with AD or dementia, with this number expected to reach 139 million in the next 50 years (Alzheimer's Disease International). Alzheimer's disease accounts for approximately 60–80% of dementia cases and is among the leading causes of death worldwide, highlighting its growing impact as a major public health concern (Alzheimer's Association, 2024). Despite extensive research, there is currently no cure for the disease, making early and accurate diagnosis essential for patient care and intervention. Many studies on AD have utilized neuroimaging techniques such as magnetic resonance imaging (MRI) to detect structural brain changes associated with the disease's progression, including brain atrophy and volume loss (Shi et al., 2025). However, some researchers have highlighted the concern that these measures such as MRIs commonly used in AD research are rather nonspecific indicators of damage to the brain that could be from a variety of other causes. (Jack et al., 2018).

Recent advances in machine learning have enabled the development of predictive models that can analyze complex biomedical data for disease classification. In particular, deep learning approaches have shown promise in identifying Alzheimer's-related patterns from neuroimaging data (Vieira et al., 2017). However, many existing studies rely on a single data modality, such as imaging alone, which may limit predictive performance. Integrating multiple data types, including both imaging and clinical variables, may provide a more comprehensive representation of disease status. This study will implement machine learning approaches to evaluate the effectiveness of multimodal data integration for Alzheimer's disease classification. Specifically, the model will be used to answer the following research question: does combining clinical and demographic variables from the OASIS-1 dataset with MRI-based features from OASIS-derived MRI slices improve classification performance compared to using either clinical data or MRI data

alone? It is hypothesized that the combined multimodal model will perform best, as clinical variables and MRI scans provide complementary information about disease status.

Methodology

The data for this study will be obtained from the Open Access Series of Imaging Studies (OASIS), collected by WashU Medicine. Specifically, OASIS-1 is a set that consists of a cross-sectional collection of 416 subjects aged 18 to 96. The OASIS-1 demographic and clinical data will be used for the structured (tabular) component of this study, while MRI scans derived from OASIS-1 and made available as preprocessed image slices on Kaggle will be used for the imaging component.

From the clinical data, the variables used will include age, sex, normalized whole brain volume (nWBV), and education (Educ). The variable Clinical Dementia Rating (CDR) will be used as the target variable with CDR = 0 indicating non-demented and CDR > 0 indicating demented or Alzheimer's related. This allows for the problem to be treated as a binary classification task.

1. **Data Preprocessing:** Before modeling, the clinical dataset will be preprocessed to remove all subjects with missing values in the selected variables. Categorical variables such as sex will be encoded and continuous features will be normalized to ensure consistent scaling across all the features. For the MRI scans, scans will be grouped by patient ID. Since each subject has multiple MRI slices, a subset of slices per subject will be randomly selected to represent each patient. These slices will be resized to a consistent resolution and normalized.
2. **Building the Models:** A total of three models will be developed and compared with each other. First, a clinical variable model will be trained using the tabular features from the OASIS dataset. This model will use logistic regression, to predict dementia status based on only clinical variables. Second, an MRI model will be constructed using a convolutional neural network trained on MRI slices of patients. Predictions will be generated at the slice level and aggregated to produce subject-level classifications. Finally, a multimodal model will be implemented by combining the two data modalities. MRI-derived feature representations from the CNN will be concatenated with the clinical variables, and the combined feature vector will be input into a multilayer perceptron classifier to generate final predictions. This architecture enables integration of imaging and clinical information into the same framework.
3. **Analysis of Model Performance:** Model performance will be evaluated using accuracy, precision, recall, F1-score, and receiver operating characteristic area under the curve (ROC-AUC). Recall will be crucial, given the importance of correctly identifying individuals with dementia. The performance of the clinical-only, MRI-only, and multimodal models will be compared to determine whether combining clinical and imaging data improves classification performance.

Expected Results

It is hypothesized that the combined multimodal model will achieve the best performance, because clinical variables and MRI scans provide complementary information about disease

status. Clinical data captures demographic and cognitive features associated with Alzheimer's disease, and MRI scans provide structural information about brain atrophy. Integrating these two modalities is expected to result in a more comprehensive representation of each patient, which can lead to improved classification compared to solely depending on one over the other. If this study confirms the hypothesis, it would demonstrate that multi-modal data capture distinct aspects of disease progression and integrating them into studies can improve overall predictive performance. Furthermore, it would highlight the potential of multimodal models to capture patient-specific characteristics, which may support the development of more personalized diagnostic approaches in future work.

A limitation of this study is that the dataset is relatively small and may contain missing clinical values, which could reduce model generalizability. Additionally, the MRI data consists of 2D slices rather than full 3D brain volumes, which may limit the model's ability to capture complete structural information. Despite these limitations, the study provides a practical framework for evaluating multimodal machine learning approaches using publicly available data.

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